SAP Professional Fundamentals — Course 1 Hands-on Project

Case Study Assignment: Tasks 1, 2, and 3 Deliverable

Customer Name:	Hitachi High-Tech Corporation (Fortune Global 500 #133 Hitachi Ltd.)
Engagement Title:	Global Cloud ERP Modernization & Clean Core Transformation
Author & Role:	Lead SAP Functional & Business Process Consultant (Solution Architect)
Target Solutions:	SAP S/4HANA Cloud (Two-Tier), SAP Analytics Cloud (SAC), SAP BTP

Task 1: Select a Case Study for Your Assignment

Customer Overview: You are Hitachi High-Tech Corporation, a major subsidiary of Fortune Global 500 company Hitachi Ltd. (ranked #133 globally). Headquartered in Tokyo, Japan, you are an admired global market leader operating across 26+ countries in high-tech manufacturing, semiconductor manufacturing equipment (etchers, CD-SEMs), clinical diagnostic instrumentation, and advanced industrial materials. Your customers include the world's most demanding semiconductor fabrication plants and leading biomedical research laboratories.

Verification Against Assignment Selection Criteria

- **Criteria 1 — Market / Industry-Leading Company:** You represent a premier global enterprise within the Fortune Global 500 ecosystem (Hitachi Ltd. #133). You command dominant market shares in critical semiconductor metrology and automated clinical analyzers, standing as a benchmark of excellence and innovation worldwide.
- **Criteria 2 — Comprehensive Digital Transformation:** You are actively executing an enterprise-wide cloud digital transformation. Rather than a superficial technical upgrade, you are undergoing a bold greenfield transformation that overhauls business processes, eliminates legacy fragmentation, and establishes an agile cloud-native operating model across all global subsidiaries.
- **Criteria 3 — Target SAP Cloud Products:** Your target solution architecture deploys the exact mandated trio of SAP cloud innovations: (1) SAP S/4HANA Cloud in a flexible Two-Tier model (Private Edition for HQ and advanced manufacturing; Public Edition for international sales and service subsidiaries); (2) SAP Analytics Cloud (SAC) for enterprise planning and live operational dashboards; and (3) SAP Business Technology Platform (SAP BTP) to execute a strict 'Clean Core' strategy with side-by-side extensibility and robust integration.

- **Criteria 4 — Rich Case Study Material & Concrete Metrics:** Your transformation is richly documented with tangible quantitative metrics—including slashing legacy custom add-on code by 94% (reducing ~9,000 custom programs to under 800 standard/BTP extensions) and standardizing core workflows across 26 countries. Furthermore, direct insights from executive stakeholders—specifically Makoto Suzuki, Senior Director and General Manager of the IT Strategy & Digital Solution Division—provide clear visibility into your strategic drivers.

Task 2: Understand Customer and Industry Context

Activity 1: Five (5) Macro Business or Technology Trends Impacting Your Industry

- **1. Geopolitical Supply Chain Realignment & Semiconductor Sovereignty:** National initiatives like the US CHIPS Act, European Chips Act, and Asian semiconductor incentives are driving localized fab construction worldwide. For you, this surge creates tremendous demand for metrology equipment alongside acute supply chain volatility, export control regulations, and the need for dynamic multi-country inventory management.
- **2. AI Acceleration & Sub-2nm Nanoscale Precision Demands:** The explosive growth of artificial intelligence and high-performance computing requires semiconductor chips manufactured at advanced sub-2nm nodes. Your manufacturing and engineering systems must ensure zero-defect quality and real-time process monitoring to support customer yield targets.
- **3. Precision Healthcare & Automated Clinical Diagnostics:** Global demographic aging and rising healthcare demands are driving hospitals and diagnostic laboratories toward high-throughput, automated clinical chemistry and immunology analyzers. This necessitates seamless cloud telemetry, automated reagent replenishment, and proactive field maintenance.
- **4. Enterprise Cloud Mandate & The 'Clean Core' Architectural Paradigm:** Across discrete manufacturing and medical technology, heavily modified on-premises ERP systems have become unsustainable bottlenecks. Market leaders are migrating to cloud-native architectures with a 'Clean Core,' allowing them to consume regular vendor innovations without multi-month regression cycles.
- **5. Global ESG Compliance & Stringent Regulatory Mandates:** Stricter international environmental regulations (EU CSRD, RoHS, REACH) and medical device quality standards (FDA 21 CFR Part 820, IVDR, ISO 13485) require granular, auditable track-and-trace capabilities across your entire product lifecycle, alongside transparent Scope 1-3 carbon accounting.

Activity 2: Two (2) Customer Strategic Motivations for Doing the Project (The 'Why?')

- **Motivation 1 — Overcoming Debilitating Legacy Technical Debt to Restore Agility:** Since implementing SAP R/3 in the late 1990s, you accumulated approximately 9,000 custom add-ons, bespoke user exits, and isolated database modifications. Over thirty years, this created severe architectural calcification: system upgrades took months or years to test, introducing new digital capabilities was painfully slow, and maintenance costs escalated. Your fundamental motivation is to escape this legacy trap through a clean-slate greenfield migration and a strict Fit-to-Standard methodology, re-establishing speed and innovation across your business.

- **Motivation 2 — Harmonizing Global Operations to Power Scalable Service-Oriented Business Models:** Decades of rapid multinational expansion left your 26+ operating entities with fragmented systems, inconsistent data models, and localized processes. Your strategic motivation is to unify these disparate organizations onto a standardized global digital backbone. This enables you to pivot from traditional one-time hardware sales to high-margin, recurring digital service models—such as Equipment-as-a-Service, predictive field maintenance, and global spare parts subscriptions.

Activity 3: Three (3) Customer Business Goals and Target Outcomes

- **Goal 1 — Drastically Reduce Total Cost of Ownership (TCO) and Cut Customizations by 94%:** Slash legacy custom code from ~9,000 add-ons down to under 800 standard/BTP-hosted extensions. By adhering strictly to standard SAP best-practice business processes out-of-the-box and isolating essential enhancements on SAP BTP, you will minimize ongoing maintenance overhead, eliminate costly upgrade regression cycles, and radically reduce IT operational expenditures.
- **Goal 2 — Accelerate Business Agility and Compress Lead Times from Months to Weeks:** Dramatically reduce the cycle time required to onboard newly acquired subsidiaries, launch new product lines, or deploy localized business capabilities. Leveraging SAP S/4HANA Cloud Public Edition for international subsidiaries and standardized APIs on SAP BTP will compress subsidiary onboarding cycles from 6–12 months down to 4–8 weeks.
- **Goal 3 — Establish Real-Time Global Visibility and Data-Driven Decision-Making Across 26+ Countries:** Eliminate multi-day financial closing delays, regional inventory blindness, and manual spreadsheet consolidation. Integrating SAP Analytics Cloud directly with live S/4HANA transactional data provides executive leadership with real-time, consolidated dashboards for manufacturing yields, supply chain inventories, field service response SLAs, and global profitability.

Activity 4: Selection of Two (2) Discovery Analysis Frameworks

Framework 1: PESTLE Analysis Model

• Key Elements: PESTLE systematically evaluates six external macro-environmental dimensions: Political (semiconductor trade policies, national fab subsidies), Economic (global R&D capital expenditure cycles, currency volatility), Social (demographic healthcare demands, skilled engineering talent availability), Technological (cloud architectures, clean core, AI sub-2nm inspection), Legal (FDA/IVDR medical compliance, export controls, intellectual property protections), and Environmental (Scope 3 carbon reduction, hazardous materials disposal, RoHS/REACH).

• Benefits for Your Engagement: PESTLE provides our engagement team with the macro-environmental intelligence necessary to architect a resilient, globally compliant solution. Because you operate across 26 countries with diverse regulatory frameworks (such as strict US export restrictions on semiconductor machinery versus EU medical device regulations), PESTLE ensures our core template configurations natively accommodate regional legal and market requirements without requiring localized ERP core modifications.

• Considerations & Gaps: While PESTLE excels at capturing external market drivers, it does not assess internal organizational friction, employee morale, or granular departmental workflow pain points. Consequently, it must be paired with an empathetic, human-centric discovery framework.

Framework 2: Design Thinking

• Key Elements: Design Thinking is an iterative, user-centric problem-solving methodology executed across five collaborative phases: Empathize (immersing in end-users' daily operational routines), Define (framing root pain points into precise challenge statements), Ideate (brainstorming standard process alternatives), Prototype (building interactive SAP Fiori and BTP mockups), and Test (validating workflows with real business users).

• Benefits for Your Engagement: Design Thinking is the single most effective methodology for overcoming your primary operational challenge: user resistance to letting go of 9,000 familiar custom add-ons. Rather than imposing standard processes top-down, we will conduct interactive Design Thinking workshops with your regional manufacturing planners, procurement buyers, and field technicians. By understanding their underlying operational intent, we can demonstrate how standard SAP Fiori apps satisfy their requirements and co-design tailored BTP side-by-side apps only where distinct competitive advantages exist, fostering genuine user ownership.

• Considerations & Gaps: Without clear boundaries, Design Thinking workshops can lead to unconstrained feature wish-lists that challenge clean core principles. Our team will tightly govern sessions within standard SAP capabilities and our Clean Core Governance Charter.

Activity 5 (Optional Stretch): Comprehensive SWOT Analysis for Hitachi High-Tech

STRENGTHS (Internal)	WEAKNESSES (Internal)
• Global market leadership in semiconductor metrology (CD-SEMs) and clinical chemistry analyzers. • Strong financial backing and R&D synergy of Hitachi Ltd. (Fortune Global 500 #133). • High brand prestige and long-standing customer trust across top tier fabs and healthcare labs. • Strong corporate leadership sponsorship for a clean core, greenfield cloud transformation.	• Severe technical debt from ~9,000 legacy custom add-ons, making upgrades slow and risky. • Siloed operations, disparate data models, and fragmented workflows across 26 countries. • Historically lengthy lead times to onboard new subsidiaries and roll out digital capabilities. • Employee change resistance when transitioning from bespoke screens to standard SAP Fiori.
OPPORTUNITIES (External)	**THREATS (External)**
• Capitalizing on surging global fab investments fueled by US/EU CHIPS Acts. • Monetizing Equipment-as-a-Service and predictive field service models via IoT and SAC. • Slashing annual IT maintenance spend by 94% to reinvest in customer-facing digital innovation. • Rapid international subsidiary expansion	• Geopolitical trade tensions, export control sanctions, and tariff shifts in semiconductor markets. • Intense competition from specialized global equipment manufacturers in Asia and the West. • Operational disruption risk to active production lines during data migration and cutover.

enabled by SAP S/4HANA Cloud Public Edition.	• Divergent regional compliance mandates (EU IVDR vs. US FDA) complicating template standardization.

Task 3: Build Your Project Team

Activity 1: Selected Project Role

My Selected Project Role: Lead SAP Functional & Business Process Consultant (Enterprise Solution Architect & Fit-to-Standard Lead)

Activity 2: Skills and Expertise Brought to the Team in This Role

- **1. SAP Activate Methodology & Fit-to-Standard Workshop Leadership:** Extensive experience directing structured Fit-to-Standard workshops in the Prepare and Explore phases. Expert at guiding business stakeholders through standard Best Practice scope items and demonstrating how out-of-the-box workflows satisfy business needs without custom coding.
- **2. End-to-End Core Business Process Re-engineering:** Comprehensive cross-functional expertise across Order-to-Cash (complex international equipment sales, warranty management), Procure-to-Pay (sub-tier component procurement, supplier collaboration), Plan-to-Produce (configure-to-order manufacturing, shop floor control), and Record-to-Report (multi-entity financial consolidation).
- **3. Clean Core Architecture & Extensibility Governance:** Rigorous ability to execute objective Delta Assessments. Skilled at distinguishing between non-differentiating operational habits (redirected to standard S/4HANA workflows) and genuine competitive differentiators (routed to SAP BTP as side-by-side low-code/pro-code extensions).
- **4. Executive Communication & Cross-Cultural Stakeholder Alignment:** Proven consulting diplomacy and negotiation skills, harmonizing differing regional requirements across Japanese corporate headquarters and diverse overseas subsidiaries in Europe and North America to secure timely consensus.

Activity 3: Other Core Project Roles, Skills/Expertise, and Justification

- **1. SAP Project Manager / Activate Engagement Lead:** Skills: PMP/SAP Activate certified, agile scrum master, global program governance, financial budget tracking, and RAID risk management.
  Why Needed: Your project spans 26 countries, two cloud tiers (Private and Public), and multiple integration points. A dedicated Project Manager ensures schedule adherence, scope control, resource coordination, and executive steering committee alignment.
- **2. SAP Technical Solution Architect & BTP Integration Specialist:** Skills: SAP BTP (Integration Suite, Extension Suite, SAP Build), ABAP Cloud RESTful Application Programming (RAP), OData/REST APIs, Cloud Connector, and enterprise cybersecurity.
  Why Needed: Slashing 9,000 legacy add-ons requires re-architecting roughly 800 strategic

extensions side-by-side on SAP BTP. This specialist designs the decoupling architecture, builds real-time API integrations between S/4HANA and boundary systems, and enforces cybersecurity.

- **3. SAP Analytics Cloud (SAC) & Enterprise Data Specialist:** Skills: Data modeling, Core Data Services (CDS) view creation, SAP Datasphere, SAC story and dashboard design, predictive planning, and live data pipelines.
  Why Needed: Achieving real-time global operational visibility requires configuring executive cockpits, automated KPI trackers, and rolling financial forecast models across all global manufacturing plants and business units without data latency.
- **4. Organizational Change Management (OCM) & User Enablement Lead:** Skills: Prosci ADKAR certification, stakeholder readiness assessment, role-based curriculum design, SAP Enable Now digital adoption, and communication strategies.
  Why Needed: Shifting from 30-year bespoke screens to standard SAP Fiori represents a profound cultural transformation. The OCM Lead builds a global Change Champion network, drives user enablement, and secures enthusiastic user adoption across all sites.
- **5. Customer Core Team Leads & Business Process Owners (BPOs) [Hitachi High-Tech]:** Skills: Deep institutional knowledge of Hitachi High-Tech's manufacturing, supply chain, and clinical diagnostic operations; authority to make binding process decisions.
  Why Needed: Outside consultants cannot succeed without business ownership. Your BPOs co-facilitate Fit-to-Standard workshops, validate business process flows, execute User Acceptance Testing (UAT), and champion new standard operating procedures across subsidiaries.

Activity 4: How Project Team Roles Will Collaborate to Deliver the SAP Implementation

Our team will operate under an agile, co-innovation model structured around the official SAP Activate Methodology across six defined phases:

• Discover & Prepare Phases: The Project Manager establishes governance charters and sprint cadences. In parallel, I (Lead Functional Consultant) collaborate with your executive sponsors and the Technical Architect to finalize scope items and provision initial cloud tenant sandboxes.

• Explore Phase: I partner directly with your Customer BPOs to conduct intensive Fit-to-Standard workshops using live SAP best practice systems. We identify operational requirements, systematically retiring non-standard customizations. When true delta requirements emerge, the Technical Architect and I evaluate them against Clean Core criteria: standard configurations are adopted where possible, while high-value competitive extensions are slated for SAP BTP side-by-side development.

• Realize Phase: The team enters two-week agile sprints. The Technical Architect builds BTP apps and API pipelines; the SAC Specialist models analytical stories; I configure core functional business processes and author test scripts; and your BPOs participate in sprint reviews, continuously validating system increments.

• Deploy & Run Phases: The OCM Lead directs global end-user training via SAP Enable Now, while the Project Manager and Technical Architect execute cutover rehearsals and data migration runs. During go-live, our cross-functional team delivers hypercare support before transitioning system stewardship to your permanent Center of Excellence (CoE).

• Clean Core Governance Board (CCGB): A weekly governance board (Lead Consultant, Technical Architect, PM, and your CIO/BPO leads) will evaluate and approve any proposed extension, ensuring your 94% customization reduction target is permanently maintained.